

STATE OF MATTER

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CONTENT

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Boyle's Law

At constant temperature, the volume of a fixed mass of gas is inversely proportional to its pressure.

The Boyle's Law may be expressed mathematically as

$$V \propto 1/P \text{ (T, n are constant)}$$

$$\text{or } V = k \times 1/P$$

where k is a proportionality constant.

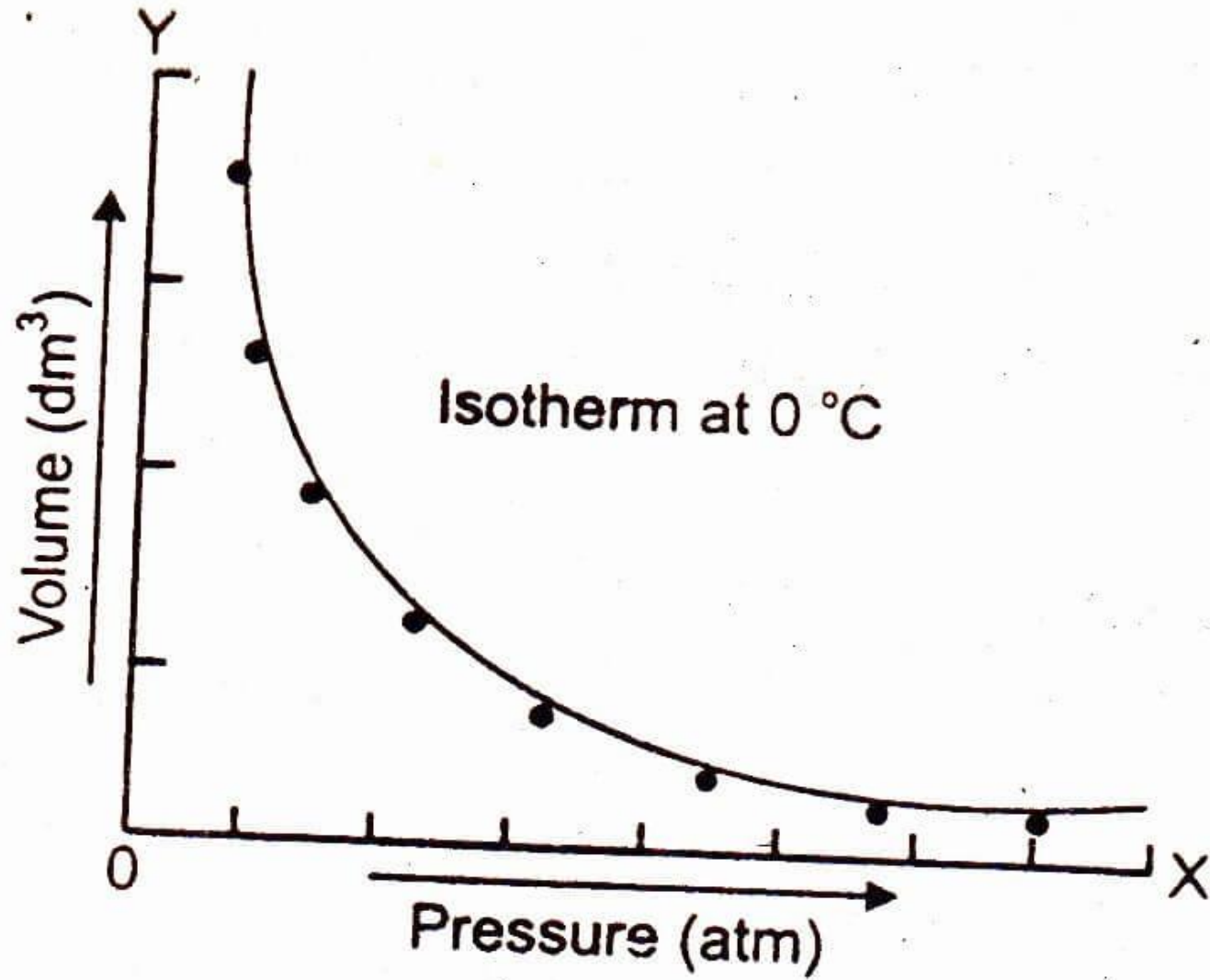
$$PV = k$$

If P_1 , V_1 are the initial pressure and volume of a given sample of gas and P_2 , V_2 the changed pressure and volume, we can write

$$P_1 V_1 = k = P_2 V_2$$

$$\text{or } P_1 V_1 = P_2 V_2$$

Fig: Boyle's Law



Charles's Law

At constant pressure, the volume of a fixed mass of gas is directly proportional to the Kelvin temperature of absolute temperature. If the absolute temperature is doubled, the volume is doubled.

Charles' Law may be expressed mathematically as

$$V \propto T \text{ (P, n are constant)}$$

$$\text{or } V = k T$$

where k is a constant.

$$\text{or } V/T = k$$

$$=k$$

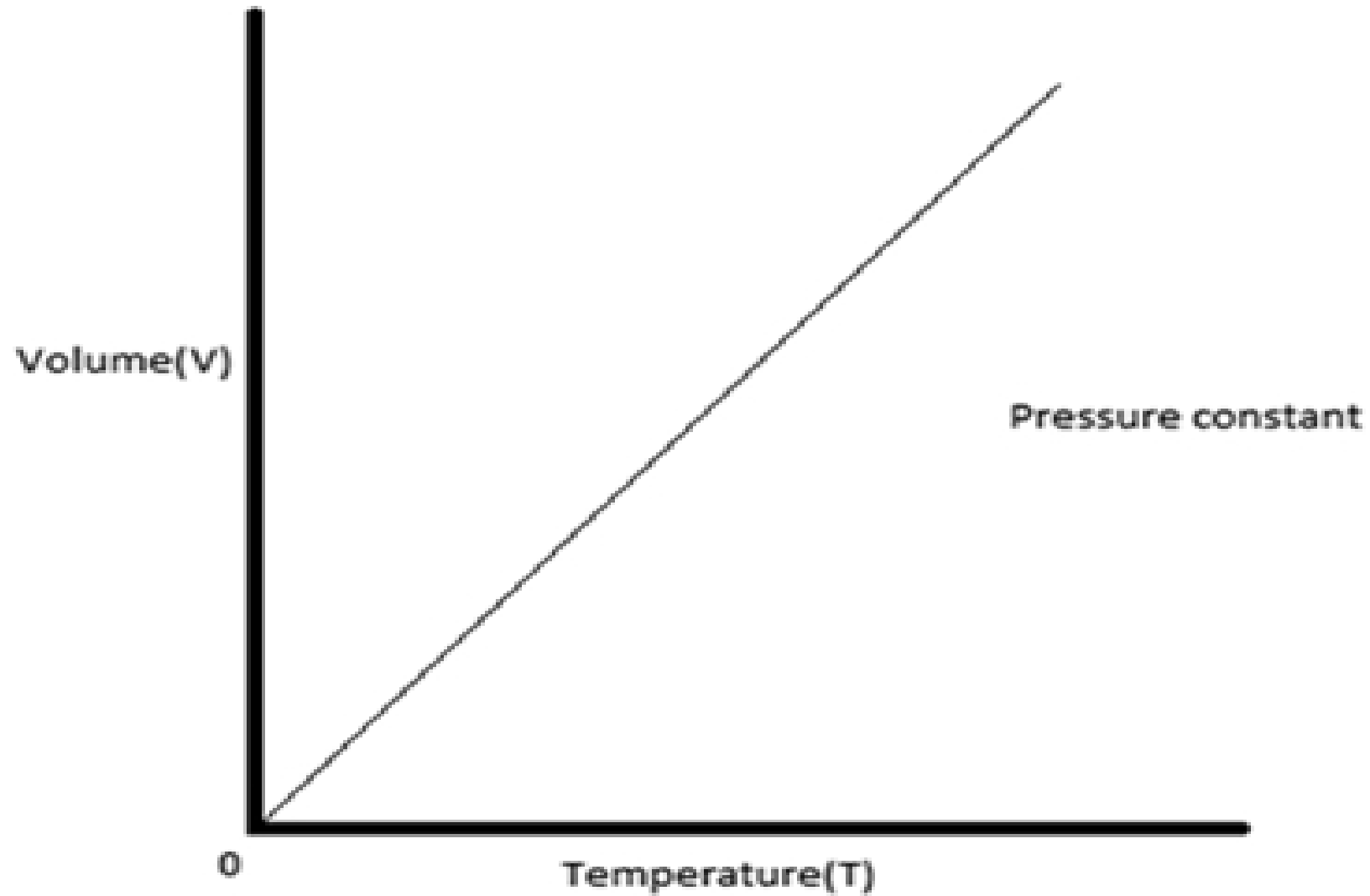
$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

T = Temperature of the gas

V = Volume of the gas

Pressure must be constant

Fig:Charle's Law



The Combined Gas Law

Boyle's Law and Charles' Law can be combined into a single relationship called the **Combined Gas Law**.

$$\text{Boyle's Law} \quad V \propto \frac{1}{P} \quad (T, n \text{ constant})$$

$$\text{Charles' Law} \quad V \propto T \quad (P, n \text{ constant})$$

$$\text{Therefore,} \quad V \propto \frac{T}{P} \quad (n \text{ constant})$$

The combined law can be stated as : **for a fixed mass of gas, the volume is directly proportional to kelvin temperature and inversely proportional to the pressure.**

If k be the proportionality constant,

$$V = \frac{kT}{P} \quad (n \text{ constant})$$

$$\text{or} \quad \frac{PV}{T} = k \quad (n \text{ constant})$$

If the pressure, volume and temperature of a gas be changed from P_1, V_1 and T_1 to P_2, T_2 and V_2 , then

$$\frac{P_1 V_1}{T_1} = k \quad \frac{P_2 V_2}{T_2} = k$$

$$\text{or} \quad \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

This is the form of combined law for two sets of conditions. It can be used to solve problems involving a change in the three variables P, V and T for a fixed mass of gas.

Gay Lussac's Law

At constant volume, the pressure of a fixed mass of gas is directly proportional to the Kelvin temperature or absolute temperature.

The law may be expressed mathematically as

$$P \propto T \text{ (Volume, n are constant)}$$

$$\text{or } P = K T$$

$$\text{or } P / T = K$$

For different conditions of pressure and temperature

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

P_1 P_2 are pressure of gas
 T_1 T_2 are Temperature of gas

Avogadro's Law

Equal volumes of gases at the same temperature and pressure contain equal number of moles or molecules. If the molar amount is doubled, the volume is doubled.

$$V \propto n \text{ (T and P constant)}$$

$$V = A n$$

where A is constant of proportionality

$$\text{Or } V / n = A$$

For any two gases with volumes
and moles at constant T and P,

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

V_1 V_2 are Volumes of gas

n_1 n_2 are amount of gas

The Ideal Gas Equation

We have studied three simple gas laws :

Boyle's Law $V \propto 1/P$ (T, n constant)

Charles' Law $V \propto T$ (n, P constant)

Avogadro's Law $V \propto n$ (P, T constant)

These three laws can be combined into a single more general gas law :

$$V \propto n T / P \dots (1)$$

$$V = R \times n T / P, R \text{ is gas constant}$$

$$PV = n RT$$

The equation is called the Ideal-gas Equation or simply the general Gas Equation.

For one mole of ($n=1$) the gas

$$PV = RT$$

Ideal and Real gas

Ideal gas: The gas which obeys the molar gas equation $PV=nRT$ at all temperatures and pressure.

Real gas: The gas which does not follow gas laws at any temperature and pressure is known as real gas. H_2, O_2, N_2 etc. do not follow Boyle's and Charles law at ordinary temp. and pressure.

At extremely low pressure and high temperature some gases particularly the noble gases behave like an ideal gas

Causes of deviation of real gas from ideal behaviour

There are two cause of deviation of real gas from ideal nature.They are

(i) Volume defect

(ii) Pressure defect

Volume defect:Based on one of the postulates of the kinetic theory of the gases is that the volume of gaseous molecules is negligibly small compared to the total volume of the gas.But practically,it is not true. Because, at low temperature and high pressure gases can liquify or solidify. These liquids and solids have certain volume, which is not negligible. Therefore, the gas equation($PV=nRT$)deduced for real gases omitting their own volume can not be obeyed by them. In the equation if the volume of the ideal gas is V then the volume of the real gas will be $(V-nb)$

Here, b = Van der Waal's constant

n = number of molecules

nb = volume of the total molecule

we know the molar volume of any gas at STP is 22.4 litre. At high pressure and low temperature real gases deviate from ideal behaviour. This deviation is explained in terms of compressibility factor, $Z = PV/RT$

Pressure defect: For an ideal gas the forces of attraction between the gaseous molecule are considered to be nil at all temperature and pressure.

After increasing pressure & decreasing temperature the gaseous molecules change into liquid due to intermolecular force. Other than this, the presence of intermolecular force has been proved from Joule Thomson effect.

Van Der Waals Equation

a) Volume Correction

- The actual volume free to move in is less because of partial size.
- More molecules will have more effect.
- Corrected volume $V' = V - n b$
- b is a constant that differs for each gas.

b) Pressure Correction

- Because the molecules are attracted to each other, the pressure on the container will be less than ideal
- Pressure depends on the number of molecules per liter.
- Since two molecules interact the effect must be squared.

$$P \text{ (actual)} = P(\text{ideal}) - n^2a/V^2$$

$$P \text{ (ideal)} = P(\text{actual}) + n^2a/V^2$$

$$\begin{array}{ccccccc}
 & & \text{correction factor to} & & \text{Correction factor to} & & \text{Temperature} \\
 & & \text{account for} & & \text{account for molecule} & & \\
 & & \text{intermolecular forces} & & \text{size} & & \\
 & \uparrow & & \uparrow & & \uparrow & \\
 (P + \frac{a n^2}{V^2}) (V - nb) = n R T \\
 \downarrow \quad \downarrow \quad \downarrow & & & & & & \\
 \text{Measure Pressure} & \text{Measure Volume} & \text{Number of Moles} & & & &
 \end{array}$$

Vander Waal Equations

$$P + \frac{a n^2}{V^2} \Rightarrow \text{Ideal Gas Pressure}$$

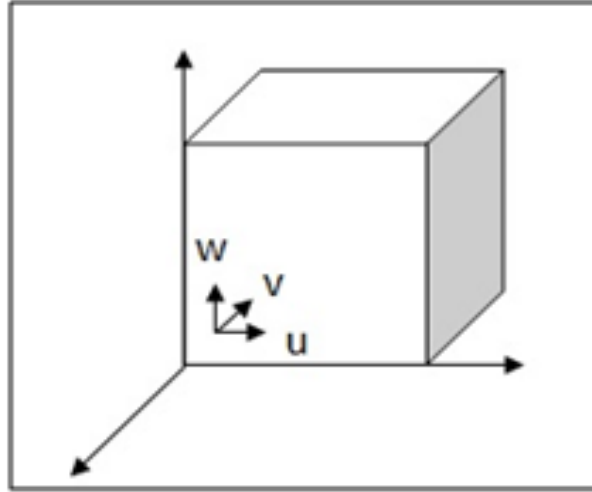
$$V - nb \Rightarrow \text{Ideal gas Volume}$$

Kinetic Theory of Gases

- 1) A gas consists of extremely small discrete particles called molecules dispersed throughout the container.
- 2) Gas molecules are in constant random motion with high velocities.
- 3) The gas molecules can move freely, independent of each other.
- 4) There is no loss of the kinetic energy of a molecule during a collision.
- 5) The pressure of a gas is caused by the hits recorded by molecules on the walls of the container.
- 6) The average kinetic energy of the gas molecules is directly proportional to absolute temperature (Kelvin temperature).

Kinetic Gas Equation

Consider a cubic container with edge length l containing n molecules of gas of molecular mass $= m$, and RMS speed $= C$ at temperature T and pressure P .



Therefore, $C^2 = u^2 + v^2 + w^2$.

Initial momentum before striking $= mu$

Final momentum after striking $= -mu$

$\therefore$ Total change in momentum $= mu - (-mu) = 2mu$

Number of collisions per sec = u/l

∴ Total change in momentum per sec per molecule on face

‘A’ = collision per second $\times$ change in momentum

$$= u/l \times 2mu$$

$$= 2mu^2 / l$$

Same as, $2mv^2 / l$, $2mw^2 / l$

Total rate change of momentum for the molecule given on the picture,
 $= 2mu^2/l + 2mv^2/l + 2mw^2/l$
 $= 2m/l (u^2 + v^2 + w^2)$
 $= 2mC^2/l$, where C = root means square velocity.

If we consider all the molecules of the gas present in this cubic container,
total change of momentum $= 2mnC^2/l$, where C = root means square
velocity.

According to Newton's second law of motion, the rate of change of momentum due to wall collision is equal to the force developed within the gases.

$$\text{Therefore, } P \times 6l^2 = (2mnC^2)/l$$

$$\text{or, } P \times 3l^3 = mnC^2$$

$$\text{or, } P \times l^3 = mnC^2/3$$

$$\text{The area of the cube} = l^2$$

$$\text{The volume of the cube} = V = l^3$$

$$PV = \frac{1}{3}mnC^2$$

Root Mean Square Speed

Root mean square speed or RMS is defined as the square root of the average of the squares of speeds of the gases.